Shivani Radhakrishnan

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EDUCATION

Northeastern University — Master of Science in Computer Science — Sep 2026 – Present

R.M.K. Engineering College — B.E., Computer Science & Engineering (Honors in AI & ML), CGPA 8.45/10 — 2022 – 2026

TECHNICAL SKILLS

Languages: Python, Swift, Kotlin, Dart

ML / AI: PyTorch, scikit-learn, Sentence-Transformers, RAG with local LLMs, Whisper, ONNX Runtime, computer vision, NLP, Core ML

Data & Cloud: Apache Spark, pandas, NumPy, AWS, Google Cloud

Apps & Tools: SwiftUI, ARKit, Create ML, Flutter, FastAPI, Docker, Streamlit, Git, GitHub Actions CI, pytest

PROJECTS

NavAssist | Swift, ARKit, Core ML, Kotlin, Python — Jun 2024 – Present

- Built a voice-first iPhone/iPad navigation app for blind and low-vision users that maps buildings as they walk and guides them back with A* routing, spoken and haptic cues and on-device OCR.
- Enabled obstacle perception on iPhones without LiDAR by converting Depth Anything V2 relative depth to meters each frame with ARKit feature points: AbsRel 0.172 → 0.070, 6 cm median error (TUM RGB-D); obstacle-zone F1 0.89–0.99 vs. 0.53–0.61 before.
- Trained a Create ML door, door-handle and stairs detector: on 717 held-out images, stairs recall 0.79 vs. 0.47 and doors 0.71 vs. 0.38 for Apple's built-in classifier at similar precision; 8.9 ms per image.
- Designed the navigation core as a platform-independent Swift package (83 unit tests) and ported it to Kotlin (79 tests), with both suites running in GitHub Actions CI on every push.

CiteVerify | Python, FastAPI, Docker, Streamlit — Dec 2025 – Mar 2026

- Built a citation-hallucination detector that resolves DOIs against Crossref, DataCite and arXiv, compares title, year and author, and falls back to title search; HTTP layer with timeouts, retries, rate limiting and caching.
- Raised F1 from 71.9% to 91.3% with 0 false positives on a 136-citation held-out benchmark that re-runs offline in CI; 89.5% vs. 54.5% F1 for the previous version on a held-out set of 200 chatbot-written citations.

EXPERIENCE

Engineer Intern — Tech Mahindra, Chennai, India (Hybrid) — Dec 2025 – Feb 2026

- Built an end-to-end NLP pipeline in Python for resume–job matching (preprocessing, Sentence-Transformer embeddings, training and evaluation) for enterprise recruitment.
- Scaled matching to 2.19M resume–job pairs by batch-encoding text and computing similarity as one matrix operation, cutting estimated runtime from hours to minutes.
- Ran bias and fairness analysis across demographic groups and documented deployment trade-offs in a technical report.

AI Intern — CodeAlpha (Virtual) — Jun 2025 – Jul 2025

- Built BotBuddy, an FAQ chatbot (sentence embeddings, scikit-learn, Streamlit), later extended with a RAG mode on a local Qwen2.5-1.5B model (llama.cpp) that cites its sources: 82.7% correct on 98 held-out questions vs. 44.9% for FAQ matching.
- Built a 100+ language translation app with service fallback, retries and caching, plus speech-to-speech via local Whisper (9.2% WER on Spanish FLEURS); pytest suite with mocked network calls in CI.

Project Intern — Virtusa, Chennai, India (Remote) — Jun 2024 – Aug 2024

- Built QuantumBuy, an accessibility-first Flutter shopping app with voice ordering, screen reading, color detection and app-wide accessibility settings, designed around users with visual, cognitive and motor disabilities.

Research Intern — Computational Intelligence Research Foundation, Chennai, India (Remote) — Mar 2023 – Apr 2023

- Developed classification and clustering models and performed exploratory data analysis on real-world datasets, presenting findings with Matplotlib and Seaborn.

PUBLICATIONS

- R. Devi, R. Shivani, R. Sathishkannan et al., “Analyzing the Performance of Collaborative Filtering in AI-Powered Recommendation Systems,” IEEE INCSST 2025. DOI: 10.1109/INCSST64791.2025.11210275. Built an ALS matrix-factorization recommender on Apache Spark MLlib (MovieLens), tuned to RMSE 0.495.

CERTIFICATIONS

ServiceNow CSA (2025) • AWS Academy: Cloud Foundations, Architecting, Security • Google Cloud Computing Foundations + 4 skill badges